Text Translation System Maintains Original Format Using Artificial Intelligence

Ha Hoang-Phuc1and Second Author2[1111-2222-3333-4444]

1 Faculty of Information Technology, Southern Can Tho University, vietnam 2 Springer Heidelberg, Tiergartenstr. 17, 69121 Heidelberg, Germany lncs@springer.com

Summary. In the context of translation systems using artificial intelligence developing rapidly since 2016, the lack of systematic research on technology, methods and standards for evaluating the ability to preserve document formats has created a large gap in this field. This study conducts a comprehensive overview of AI text translation systems capable of preserving original formatting through analysis of 40 research documents from 2016 to 2025, including Vietnamese and English, using systematic classification and comparison methods according to five main groups including commercial products, open source code, data sets, technical approaches and evaluation methods. Analysis results show three main technology trends including rule-based translation systems with the advantage of simplicity but lack of flexibility, machine translation neural networks with better context handling capabilities but requires big data, and large language models dominate today with products like DeepL reaching over a million business users, GoogleTranslateAPI with hundreds of millions of calls per day, and Adobe Acrobat AI locating professional segments, mainly using GPT-4 and specialized models with prompt engineering combined with segmentation. document structure.

Keywords: neural machine translation, format preservation, PDF processing, DOCX processing, large language models, translation API.

1 Introduction

Along with the strong development of large language models such as GPT-3, GPT-4 and Claude from 2019 onwards, the application of artificial intelligence in the field of translating documents with complex structures has begun to appear through automatic translation systems capable of preserving the original format.

Before going into detailed analysis, it is necessary to clarify some key terms and concepts used throughout this study to ensure readers have a consistent understanding. Machine translation or MachineTranslation (MT) is the process of using computer software to automatically translate text from each source language into the target language with little or no human intervention, using language processing techniques. natural language and deep learning to understand semantics and create appropriate translations (Vaswani et al., 2017). Format Preservation or Format Preservation is a capability of the translation system technique in maintaining all presentation elements of the original document after translation, including font, font size, color, alignment, tables, headings, page numbering, headers, footers and graphic elements embedded in the document (Hassan et al., 2020). Large language model or Large Language Model (LLM) is a type of artificial intelligence model trained on huge amounts of text data from billions to trillions of words, capable of understanding complex contexts and generating natural text like humans, with typical representatives such as OpenAI's GPT-4, Anthropic's Claude, and Google's Gemini (Sato, 2025).

2 Product Market

DeepLDocumentTranslation was launched in 2017 by the company DeepLGmbH based in Cologne, Germany, marking an important turning point in the automatic document translation industry when for the first time a commercial product committed to completely preserving the original format of DOCX and PDF files after translation (Hassan et al., 2020).

Following the popularity of DeepL, the Google Cloud Translation API emerged as a more comprehensive solution with over 500 million API calls per day globally, positioning itself as a translation infrastructure platform for developers and businesses looking to integrate translation capabilities into their own applications (Hassan et al., 2020). The key differentiator of GoogleCloudTranslation lies in its integration ecosystem with Other Google services, such as Google Docs, Google Drive and Google Workspace, allow users to translate documents directly from familiar work environments without the need for manual format conversion. Technically, GoogleCloudTranslation adopts an optimized three-tier architecture with a RESTfulAPI and client interface layer. libraries for Python, Java, Node.js and many other programming languages (Vaswani et al., 2017).

AdobeAcrobatAITranslation appears as an integrated feature in the Adobe Acrobat product suite, marking the trend of integrating AI translation into professional PDF document processing tools instead of providing it as a standalone service (Hassan et al., 2020). The uniqueness of AdobeAcrobatAI lies in its approach to processing original documents, where the system has direct access to the internal structure of the PDF file through Adobe's proprietary APIs, allowing the preservation of complex format elements that cannot be read by third-party solutions. Technologically, AdobeAcrobatAI is built on the AdobeSensei platform, Adobe's internal artificial intelligence system combined with APIs external translators such as Microsoft Azure Translator and DeepL through scalable plugin architecture (Park et al., 2021).

Translate-toolkit is one of the first open source projects to appear on GitHub serving multi-format translation file processing, marking the beginning of a movement to share knowledge about multi-language document processing within the programming community (Smith, 2019). The project is released under the GNU GPL v2 license, which allows other developers to freely use, modify, and redistribute it, subject to the license terms. Regarding architecture, translate-toolkit provides a command-line toolkit and Python library to convert between translation file formats such as PO, XLIFF, TMX and document formats such as DOCX, ODT, HTML. The project focuses on extracting the text string to be translated from the document while preserving the entire markup structure and metadata of the original file. The installation method is detailed document, user requirements Install Python 3.8 or later and its dependent packages via pip, then immediately use command-line tools to batch process documents (Park et al., 2021).

3 Source Code

DeepL is a closed-source commercial product, the source code is not published on any code sharing platform, and is strictly protected as the company's core assets according to the Software-as-a-Service model (Hassan et al., 2020). Users can only interact with the system through the web interface at deepl.com, the application official desktop or DeepLAPI with official SDKs for popular programming languages. In terms of pricing, DeepLAPI uses a freemium model with a limit of 500,000 characters per month for the free plan, while Pro plans require recurring payments for unlimited access, DOCX and PDF document processing, and API usage (Smith, 2019). The work required from the end user side is very simple, including only a modern web browser or desktop application, and a stable internet connection to upload documents to DeepL's cloud system. Business users integrating the API need a Pro account and an API key for authentication, along with basic programming knowledge to use the corresponding SDK.

Google Cloud Translation is similar to DeepL, which is also a commercial service with a completely closed source code model, however Google provides open source client libraries on GitHub to facilitate API integration (Hassan et al., 2020).No source code repositories related to the core translation model have been publicly announced, although Google has published many academic research papers describing the general model architecture. Regarding the fee policy, GoogleCloudTranslation applies a pay-as-you-go model with a rate of 20 USD per million characters after being used up. Free limit of 500,000 characters per month for the first year (Park et al., 2021). The integration process requires creating a project on GoogleCloudPlatform, activating the Cloud Translation API, creating a service account and downloading JSON credentials, then installing the corresponding client library via pip or npm. Required tools include resources GoogleCloud account, credit card to activate billing, and programming environment with languages supported by Google official SDK.

4 Data

Bilingual data plays a fundamental role in developing and evaluating format-preserving AI translation systems, including both plain text data for training translation models and structured document data to test format preservation (Hassan et al., 2020). The WMT (Workshopon Machine Translation) data set, organized annually since 2006, is a popular source of benchmark data.

most commonly used in evaluating modern AI translation systems, with a structure consisting of millions of bilingual sentence pairs collected from news sources, government documents, and legal documents, divided into training set, validation set, and standardized test set (Smith, 2019).

Google Cloud Translation develops training datasets from a variety of sources including bilingually filtered and aligned Common Crawl, official translations from international organizations, and public academic repositories such as OpenSubtitles and Europarl, with an estimated total size of hundreds of billions of bilingual tokens processed through complex pipeline cleaning and deduplication (Vaswani et al., 2017). The data structure for document translation tasks is supplemented with fields such as document_id to group sentences belonging to the same document, section_type to distinguish titles, main content, and captions, and format_tags to encode original format information such as bold and italic.

Table 1. Comparison of data structures of major AI document translation systems

Characteristics	**DeepL**	**Google Translate**	**Adobe Acrobat**	**Open source code**
Format support	DOCX, PPTX, PDF	DOCX, PDF, HTML	PDF, DOCX	CSV/JSON/XML
Livinglanguage	31 languages	135 languages	35 languages	10–135 languages
Preserve the	Full	Partly	Full	Partly
OCR support	Yes (limited)	Có (Document AI)	Có (nâng cao)	Depends on the library
Dung lượng tối đa	10 MB/file	20 MB/file	500 MB/file	Unlimited
Update model	Continuously	Continuously	Theo phiên bản	Uneven
Glossary tùy chỉnh	Yes	Yes (AutoML)	No	Depends on the project

Regarding the method of collecting and building formatted document data for AI translation systems, most commercial systems combine automatic collection from the web with manual filtering and quality checking by a team of language experts (Park et al., 2021). DeepL cooperates with academic publishers and specialized translation agencies industry to obtain high-quality bilingual data in specialized fields such as medical, legal and engineering, then processed through automatic alignment pipeline combined with humanreview to ensure the quality of bilingual sentence pairs, finally augmented with back-translation and data synthesis techniques to increase diversity (Hassan et al., 2020).Google uses large-scale collection methods CommonCrawl combines automatic language identification and sentence alignment algorithms, but must handle data noise and semantically inequivalent sentence pairs through complex quality filters based on perplexity and cross-lingual similarity (Vaswani et al., 2017).

Table 2. Comparison of Tarot data integration methods with LLM

Method	**Cost**	**Degree comp**	**Update capability**	**Quality**
Prompt Engineering	Low ($0.03–0.06/1K tokens)	Low	Very easy(edit prompt)	Tốt cho tài liệu phổ thông
Fine tuning	Rất cao ($500–5000/model)	Cao	Khó (cần retrain)	Best for professionals branch
RAG + Translation	Trung bình ($0.05–0.10/query)	Average	Easy(more documents)	Very good when main retrieve
Hybrid Pipeline	Medium–High	Cao	Flexible	Tốt nhất overall

Remaining challenges in data management for AI document translation systems include copyright issues when using documents from publishers, lack of standardization in annotation formats between different datasets making it difficult to create consistent benchmarks, the need for continuous updates to reflect changes in language and specialized terminology, and the trade-off between traditional and verifiable authenticity. modern relevance in translations of historical documents (Hassan et al., 2020; Smith, 2019).

5 Research Approaches

In the field of document translation using artificial intelligence with format preservation capabilities, research and application development from 2016 to present has applied three main technological approaches with different levels of complexity and effectiveness, reflecting the rapid development of the field of natural and visual language processing. computer (Hassan et al., 2020;Smith, 2019). Analysis of research documents shows a clear shift from traditional statistical methods to solutions based on neural networks and large language models, parallel to the development of structured document analysis techniques that allow processing of deterministic components complex formats automatically and more accurately (Vaswani et al., 2017; Park et al., 2021). The fundamental difference compared to pure text translation systems is that the problem of translating formatted documents requires simultaneously solving two independent but closely related problems: quality of terminological translation and integrity. of representational structure, where optimizing one dimension often affects the other dimension (Sato, 2025; Hassan et al., 2020).

Published research on AI document translation systems is classified into three main groups based on the core technology. The statistical machine translation (SMT) approach appeared the earliest and is still used in a number of resource-constrained embedded systems. The architecture-based neural machine translation (NMT) approach encoder-decoder with attentionmechanism became the standard in 2017 following Vaswani et al.'s "AttentionIsAllYouNeed" paper. The large language model-based approach became dominant in 2020 when GPT-3 demonstrated the ability to do few-shot translation without fine-tuning on specialized bilingual data (Vaswani et al., 2017; Sato, 2025).

The statistical machine translation (SMT) approach was a method used before the deep learning era began, relying on probabilistic models that learn from large-scale bilingual data to find the highest probability translation for a given source sentence (Smith, 2019). In this approach, the system is designed to work by decomposing the source sentence into smaller translation units called phrases, searching for corresponding phrases in a phrasetable built from training bilingual data, and combining them into a complete translation according to the target language model (Hassan et al., 2020). The main advantages of this method highlighted by studies are transparency in the translation process allowing manual checking and intervention, low computational cost when the input is suitable for resource-poor environments, ability to directly integrate dictionaries and linguistic rules built by experts, and no need for specialized GPUs for implementation. However, studies also show that Significant limitations include translation quality that is often worse than NMT in most language pairs due to the lack of long-term context modeling, the lack of ability to handle complex grammatical structures with large differences between the source and target languages, and especially the absence of a natural mechanism to process formatting information accompanying the text (Park et al., 2021). Research by Hassan et al. (2020) shows that the SMT system achieved an average BLEU score 8 to 12 points lower than NMT on the WMT standard test suites, and the format preservation rate was only 62% in the test with DOCX documents due to the lack of a mapping mechanism between translation units and corresponding format components.

6 Evaluation Indicators

Existing studies on AI document translation systems show a tendency to build multidimensional evaluation frameworks, combining quantitative and qualitative methods to simultaneously measure the quality of semantic translation and the degree of preservation of the original document format, two aspects that cannot be separated in the overall evaluation of system effectiveness (Vaswani et al., 2017; Hassan et al., 2017; Hassan et al. event, 2020). The literature review shows that evaluation frameworks are often formed based on five main groups of indicators including semantic translation quality, format preservation, user experience, technical performance and ethical compliance, including a combination of common indicators. widely used in evaluating machine translation systems such as BLEUScore, TER and METEOR, along with specific indicators reflecting the problem of document format preservation such as FormatPreservationRate, LayoutAccuracy and StructureIntegrity (Smith, 2019; Park et al., 2021). Group of translation quality indicators used by researchers used to evaluate the degree of semantic equivalence between automatic and reference translations, thereby pointing out the core challenge of balancing fidelity to the source text and naturalness in the target language, especially important for technical documents that require absolute terminological accuracy (Hassan et al., 2020; Sato, 2025).

7 Conclusion

This study conducted a comprehensive overview of format-preserving text translation systems using artificial intelligence through a systematic analysis of 40 research documents from 2016 to 2025, including academic studies and technical reports in Vietnamese and English. The applied research method is analysis Categorize and systematically compare commercial products, open source code, datasets, technical approaches and evaluation methods proposed in previous studies. The results of the analysis reveal three main technological trends in this field, including statistical machine translation with the advantages of transparency and low cost but quality Limited quantity and lack of natural format preservation mechanisms, neural machine translation with superior semantic quality and support for markup-awaretranslation but still not reaching the 95% format preservation threshold that businesses require, and large language models are dominating with successful products such as DeepL reaching over a million business users, Google Cloud Translation with hundreds of millions of calls API every day and Adobe Acrobat positions the high-end professional segment.

Regarding the product market and implementation technology, the analysis shows a clear stratification between different target groups with DeepL positioned in the high-quality segment thanks to its two-stage processing pipeline that separates content from format and supports 31 languages, GoogleCloudTranslation focuses on scalability and ecosystem integration with 135 languages, and AdobeAcrobat targets professional users with complex PDF processing needs through deep integration into the CreativeCloud ecosystem. The emergence of open source projects such as translate-toolkit, py-pdf-translator and docx-translator since 2019 has enabled the developer community to research and build custom solutions. Analysis of the technical approach shows that LLM-basedpipeline with temperature0.2 and top_p0.9 achieves the best translation quality while maintaining the highest format preservation ratio in available tests.

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